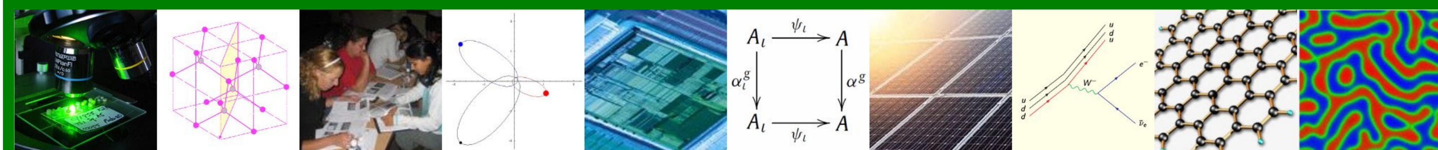
Faculty of Natural & Agricultural Sciences

Department of Physics

FSK 116: General Physics for Engineers

Version 2025.1



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Contents

1 Welcome

Welcome to FSK 116 2025.

Download this study guide and keep it handy as a reference source.

Some good news first - you will be familiar with many of the topics covered in this course. You are likely to have encountered them from everyday life experiences and from what you have learnt at school or elsewhere. This is so because Physics is at the heart of everything in nature!

Mathematics is a big part of Physics! It is the language of science. Because you will be beginning to practice Physics at an advanced level, the formalisms are stricter: the mathematics applies more generally and the definitions employed/required are more precise. Therefore, you will be treating familiar concepts using more powerful formulations and techniques.

Thinking logically is a critical requirement in the course. You will encounter situations where the final answer to a problem you are working on is correct, but because the intervening steps are not logical or are somewhat lacking, full marks are not awarded! How you get to the final answer is much more important than simply getting the correct final answer! At times, the logic will be so fatally flawed that we cannot continue marking the problem beyond the fatal error!

Lastly, the prescribed textbook is excellent! It is beautifully written and quite enjoyable to read. So, make it your companion; and don't dispose of it after passing your first year; you will need it as a starting guide to advanced topics in future!

The University of Pretoria is moving towards an "flipped classroom" approach wherein the student takes a much more active role in his/her learning. You are required to engage with the prescribed work before coming to class and to engage with your peers, tutors and with the lecturer between classes. The course rests on the three pillars: the lectures, the weekly practicals and the weekly tutorials sessions.

Because of the large number of students involved, we will communicate using strictly defined channels and modes: mainly ClickUP, various Google Forms, internal emails and the WebAssign platform.

Please study the rest of this Study Guide and browse the content of the FSK 116 ClickUP page. These contain very important information about the course.

All the best for your studies this semester.

Quintin Odendaal

2 Organisational Component

2.1 Contact Information

Mr RQ Odendaal
Office 5-51, Natural Science 1 Building (NS1)
Email: fsk116@up.ac.za
Consulting Hours: Check ClickUP or email me for an appointment.

Mrs D Bonner (Course Administration)
Office 4-11, NS1 Building
Email: fsk116.admin@up.ac.za

Mr Mufaro Makonese (Practical co-ordinator)
Email: fsk116.pracs@up.ac.za

In all correspondence regarding the tutorials and practicals, please include FSK 116, your tutorial group number and day, in subject line of your email.

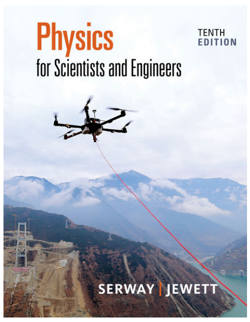
2.2 Prescribed Textbook

It is essential that you have access to a copy of the prescribed textbook as it will be referred to during the lectures and all tutorial problems will be assigned from it.

Each registered FSK 116 student will be given access - through Cengage's WebAssign platform - to an electronic copy of the prescribed textbook for the duration of the course. It is however, much better to have a print or electronic copy in your personal library as a reference textbook. This is an excellent book to have for future reference.

The prescribed textbook is:

R.A. Serway and J.W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, Published by Brooks/Cole (Cengage), 2019 10th Edition ISBN-13: 978-1-337-55329-2.



The textbook also contains a number of *worked out Examples*, some of which will be discussed in the lectures. Carefully work through the *worked out Examples* since they guide you through the solution.

2.3 Modes of presentation

There are three main modes of presentation/contact of this course, namely:
Lectures: 4 × 50 minute Lectures a week
Tutorials: 90 minutes a week
Practicals: 90 minutes a week

Every week, there is a composite 3 hour Tutorial + Practicals session, which is divided into two 1.5 hour sections.

2.3.1 Lecture Times

Contact sessions: There are 4 lectures per week. There are two lecture groups - in-person lectures are **not** repeated or recorded. The venues for the lectures is given below.

Lectures for Group 1: Mechanical and Computer Engineering

Day	Time	Venue
Monday	13:30 - 14:20	A E du Toit Auditorium
Tuesday	12:30 - 13:20	A E du Toit Auditorium
Thursday	09:30 - 10:20	A E du Toit Auditorium
Friday	12:30 - 13:20	A E du Toit Auditorium

Lectures for Group 2: Industrial and Chemical Engineering / Industrial, Mechanical and Computer Engineering for Engage students

Day	Time	Venue
Monday	09:30 - 10:20	A E du Toit Auditorium
Tuesday	08:30 - 09:20	Thuto 1-2
Thursday	07:30 - 08:20	A E du Toit Auditorium
Friday	10:30 - 11:20	Thuto 3-2

The lectures are in-person sessions. Attendance and participation during scheduled lectures periods forms an integral part of the course and important information is disseminated during these times.

During the scheduled contact (lecture) sessions, we will deal with the some of the important theoretical concepts and the derivation of useful equation to use in calculations. A set of lecture notes will be made in class. The application of the theory will be shown using example problems.

2.3.2 Practicals and Tutorials

One 3 hour session is scheduled each week for tutorials and practicals. This session is split between a practical session (90 minutes) and a tutorial session (90 minutes). See the University of Pretoria 2025 Timetable for times allocated to your specific course.

For FSK 116, there are two practical/tutorial sessions: The first is on Tuesday mornings from 10:30 to 13:30 (Group 2) and the second session is on Wednesday afternoons from 14:30 to 17:30 (Group 1).

The practical and tutorial sessions are **COMPULSORY**. Non-attendance will lead to **exam refusal** in the case of practicals.

Should you be repeating the course having done either FSK 116 or FSK 176 previously, AND were admitted to the exam, AND obtained > 50% for your practical component, you **MAY** qualify (**not** will qualify) for **PRACTICAL exemption** (this is not automatic). Please see the practical folder on ClickUP for more information on how to apply for PRACTICAL exemption.

There is **NO tutorial exemption** in this course - **tutorials are compulsory for ALL registered students**.

Venues and smaller practical/tutorial groups will be posted on ClickUP. Practicals and tutorials are compulsory.

2.4 ClickUP

ClickUP is our primary method of communicating with you. Please make sure that you login to ClickUP every few days. All important information, such as occasional extra class notes, tutorial and homework problems, marks, practical schedules, practical group allocations, test dates and venues etc. will be posted on ClickUP. Ensure that you check your TUKS email account daily (or forward your TUKS email to your preferred email address) since all correspondence will be sent to your TUKS email address by default from ClickUP. This is not just for Physics, but all your courses.

2.5 Important Dates

Lectures Start:	10 February 2025
Tutorials Start:	Week of 17 February 2025
Practicals Start:	Week of 17 February 2025
Online Tests:	Dates will be announced on ClickUP and in class
Semester Test 1:	26 March 2025
Practical Test 1:	Week of 17 March 2025
Semester Test 2:	6 May 2025
Aegrotat (Sick) Test:	22 May 2025 (provisional)
Practical Test 2:	Week of 26 May 2025
Lectures end:	29 May 2025
Examination:	6 June 2025 (provisional)

2.6 Final Marks

Final marks will be published by the faculty administration and will be available on your UP Portal page. Do not use "Print/View Academic record" to obtain your results.

The Physics Department may not release any final marks to students. Below are some codes you may encounter when obtaining your final mark.

984	Did not comply with minimum requirements (e.g. do not meet minimum practical requirement)
987	Absent from exam
988	Exam refusal
989	Exam result not available
992	Subminimum not obtained in exam (exam mark $< 40\%$)
999	Admitted to supplementary exam
994	Admitted to aegrotat (sick/special) exam

2.7 Assessment Policy

2.7.1 General

Physics cannot be mastered in a few days of "cramming". Physics needs to be practiced in order to improve your problem solving and application skills. The course credits are 16, and therefore you should spend about 160 hours on this course during the semester. This equates to spending about 9 hours per week on Physics. This will include preparing and attending the lecture sessions, consolidating the theory, preparation for and attending the tutorials (homework problems), and preparation for, doing the practicals and writing up the report.

In order to encourage students to study throughout the semester, a strong emphasis is placed on continuous assessment. For this reason the semester mark (made up of semester tests, online tests, tutorials tests and practicals conducted during the semester) makes up half of the final mark.

Below is a summary of the various assessment opportunities and the table shows the weights associated with each to make up your semester mark.
You will be able to see the cumulative average of the different assessments during the semester in Gradebook. The mark will build up as the semester progresses.

Assessment	Weight
Semester Test 1	25%
Semester Test 2	30%
Online Test Average	10%
Weekly Tutorials	15%
Weekly Practicals	10%
Practical Test 1	5%
Practical Test 2	5%
Semester Mark	**100%**

2.7.2 Assessments

NO PROGRAMMABLE CALCULATORS MAY BE USED DURING THE SEMESTER TESTS OR EXAMS.

- **Online Tests:** The online tests will either be a ClickUP test or a WebAssign tests. The ClickUP announcement will make it clear which type of test it is. Online tests will take the place of class tests, and will be made available on ClickUP during the semester. An announcement will be placed informing you of the period during which the test will be available. These tests contribute a significant amount (10%) to your semester mark.
  These tests will be available after a section of work has been completed. The online tests will be available for a two days and have a definite strict opening date and time and closing date and time. These will also be timed assessments. These tests can be completed anywhere, so you do not have to do them in the IT labs on campus - you just need internet access and a laptop or PC. It is not advisable to complete these tests on your smart phone **It is strongly advised that you take a screen shot of the question and your submitted WebAssign answers before submission** in-case there is a problem when saving your answers to WebAssign.

- **Tutorial tests:** Tutorial tests will be written at the END of each tutorial session. These tests will be used in the calculation of your Semester mark. These tutorial tests will be written during the last 15 - 20 minutes of the tutorial session. The average of all your weekly tutorial tests will make up 15% of your final semester mark.

- **Semester tests:** These tests will cover a larger amount of work than the Online Tests and will be held during the allocated Test Weeks which are scheduled in the timetable (See important dates). Details of the scope will be announced on ClickUP. The two semester tests make up a total of 55% of your semester mark.

- **Practicals:** Each week there is a practical session and each week you will complete a practical report during the allocated time. You will submit your report book to the demonstrator after the session for marking. The weekly practicals make up 10% of your semester mark.

- **Practical tests:** Two practical tests will be written during the semester and will cover the basic concepts of work done during the practical sessions.

Practical Test 1 will cover the work done during the first three practical session that discuss graphing of data, and measurement techniques.You will be given the data, and will then have to show that you understand how to draw a graph, and from the graph you will have to calculate certain values or interpret a given graph.

Practical Test 2 will cover aspects of the four experiments that you completed during the semester.

The preliminary dates are listed in paragraph 2.5 "Important Dates" above and more details will be announced on ClickUP closer to the time. The practical tests contribute 10% towards your semester mark, making the practical component of you semester mark 20% !

2.7.3 Absent from a scheduled Assessment

Should you have been ill for any of the assessments listed above, you will need to complete the Google form "Absent from Assessment" which you can find in the "Forms" folder on ClickUP. The form makes provision for you to **submit proof** of why you did not submit/attend the assessment e.g. doctors certificate, load shedding schedule for your area, connectivity issues etc. It is essential that you submit the Google form as soon as possible (Time-stamp should be within 1 week of the missed session). **Should you not timeously submit a valid reason on the Google form** along with the relevant documentation, you will receive zero (0) for that assessment.

Only medical certificates that comply with the University regulations will be accepted. (Doctors need to have been consulted <u>on</u> the day of the assessment or a day or two <u>before</u> – *NOT* THE DAY <u>AFTER</u> THE ASSESSMENT! !)

The following guidelines will be followed:

- **Tutorial tests:** There will not be an additional tutorial test. Should you have a valid reason (recognised excuse by the University) for not submitting the tutorial test, the average of the remaining tutorial tests will be taken to calculate your semester mark.

- **Weekly practicals:** There will be opportunities to do a catch-up practical during the semester, so keep an eye on the announcements. **Failure to register and attend these sessions** will result in you receiving zero (0) for the specific practical, and this can jeopardize your exam entrance requirement for the course.

- **Online tests:** There is no sick test for the online tests. A valid reason for not being able to submit the online test during the time it was available must be submitted. Submitting a valid reason for missing the online test will result in that specific online test not contributing to the semester mark. During the semester you will have to complete at least one online test, or else receive zero (0) for this component of your semester mark.

- **Semester tests:** If you were not able to write the semester test during the test week, and you complete the Google form "Absent from Assessment" with a valid reason, you will be admitted to write the Sick Test.
  One aegrotat (sick) test has been be scheduled near the end of the semester (See Section 2.5

Important Dates above) and this will cover **ALL** the work covered during the semester up to the week before the sick test, not just the section that was covered in the semester test that you missed. This will be the only test for students who missed a semester test during the semester. The provisional date for the test can be found in section 2.5 and will be confirmed closer to the time in a ClickUP announcements and on the calendar.

- **Examination:** Should you be ill or unable to write the examination with a valid reason, you have to apply for an aegrotat examination via the **faculty administration not the Physics Department**. You can inform the department that you were ill and did not write the exam using the Google form "Absent from Assessment", but this is ***NOT* the application process to write the Aegrotat Exam**.
  It is important to note that if you were not feeling well on the day of the exam, but you sat for the exam, you will not qualify to write the Aegrotat (sick) Exam. It is advisable that if you are not feeling well, you should not attempt to write the exam but rather consult a doctor and apply to write the Aegrotat Exam.

2.7.4 Examinations

- **Admission to Examination:**
  NB: In order to be admitted to the examination, the following criteria need to be met:

  1. Semester mark of minimum 40%.
  2. Final practical mark (combination of weekly practicals and practical tests) of a minimum of 50%.
  3. Attendance of 90% of the practicals and tutorials (you need to do 6 out of the 7 practicals).

  You need to meet ALL 3 of the requirements listed above. Failure to meet **ANY** of these criteria will result in refusal to write the examination.

  Please see the regulation Eng. 2 (b) in the Yearbook of the Engineering Faculty.

- **Aegrotat (Sick) and Supplementary Examinations:**
  Aegrotat (Sick) Examination: Should you have been ill for the examination, you have to apply for an aegrotat (sick) examination via the faculty administration. The Aegrotat Examination usually written on the same day as the Supplementary Examination, but in a different venue.

  Supplementary examinations are granted automatically when one of the following criteria apply (See EBIT Yearbook section 2 (g)):

1. A final mark of between 40% and 49% has been obtained in first-year modules in the first semester
2. A pass mark has been obtained, but the required subminimum in the examination (40%) has not been obtained.

The supplementary exam will be written on the date published by the University in the supplementary exam timetable. Special supplementary examinations will not be arranged for students who were not able to write the supplementary examinations during scheduled times, as given in the examination timetable.

NB: *The maximum final mark for the module which can be awarded after supplementary examination is 50%.*
This is in accordance with regulation Eng. 2 (g) in the Yearbook of the Engineering Faculty.

2.7.5 Pass Mark

A minimum FINAL MARK of 50% AND an examination mark of at least 40% must be obtained in order to pass this module.
NB: Even if a student obtains 50% or more for the final mark, BUT obtains less than 40% for the examination, that student will not pass the module, but will be granted a supplementary examination.
NB: A student will pass the module with distinction if the final mark is 75%.

This is in accordance with Regulation Eng. 2 (j) in the Yearbook of the Engineering Faculty.

2.7.6 Cheat Sheet

During semester tests and exams, NO FORMULA sheet will be handed out with the question paper. It will be your responsibility to draw up your own formula sheet — your Cheat Sheet.

Rules for the Cheat Sheet:

- Read the instructions at the top of the Cheat Sheet.
- It must be on the official Cheat Sheet for the specific test/exam printed on A4.
- It must be hand written.
- It must be written in colour – any colour but NOT BLACK.
- It may NOT have derivations or worked out solutions to specific problems.
- It must be handed in with your test. Please make a copy of it if you would like to refer to it again later in the semester.
- The page may not be folded.
- The cheat sheet is part of your preparation for the test or exam, and you will receive an additional 2% for handing in your cheat sheet AND having your barcode sticker - see Section 2.7.7 below. If you do not hand in your cheat sheet, you will not receive the 2%.
- You may only have **ONE** cheat sheet in the exam/test venue i.e. one single sheet of paper.
- You may only write on one side of the page for a semester test, but for the sick test or the examination, you may use both sides of the page.

2.7.7 Barcodes

You will receive a set of sticky label barcodes which you must stick on everything that you submit, including all your tests, practicals, exam etc. If you misplace your barcodes, you will have to request a replacement set at a cost of R5. Keep your sheet in a safe place where it will not get worn out or torn - a plastic sleeve works well. If you do not stick your barcode onto your tests/exam, you will not receive the additional 2% bonus discussed in Section 2.7.6.

2.7.8 Grading and administrative errors

When your assessments are returned to you, please ensure that all the questions have been marked, the marks allocated on your test are correct and that the **correct marks** appear on ClickUP. Should there be any unmarked questions, these must be brought to our attention **immediately**. Should there be any other discrepancies, please complete the "Remark of semester test" (on ClickUP in the Forms folder) within 7 days of the marks and test papers becoming available. **NO REMARKS WILL BE**

DONE IF REQUESTS ARE RECEIVED LATER THAN ONE WEEK AFTER THE GRADED PAPERS HAVE BEEN MADE AVAILABLE. There will be NO remarking of scripts **at the end of the semester** to improve your semester mark.

2.7.9 Practical part of the course

ATTENDANCE OF PRACTICAL SESSIONS IS COMPULSORY

All students will be placed into a smaller group for tutorial and practical sessions. The allocated groups will be announced on ClickUP.

NB: Should you not be on the practical list – please inform Mr Mufaro Makonese immediately! EVERYONE has to attend the practical sessions.

You have to attend all practical sessions. If you have missed a session with a valid excuse (there are not many of these), you need to register for a catch-up lab by completing the Google form "Absent from Assessment" which you can find in the "Forms" folder on ClickUP. A maximum of two catch-up sessions will be held during the semester.

Students who have previously done the FSK176 or FSK116 course, **may apply for exemption** from the practical course. Two of the criteria for exemption of practicals ONLY is that you needed to have **passed** your previous practical component (50%) **AND** you need to have been admitted to the examination. Successful applicants will be notified, but **until such notification has been received you need to attend the practicals**.

Please note that successful applicants still have to attend the *COMPULSORY TUTORIALS*.

Please note: A practical mark of at least 50% and attendance of at least 90% of practical sessions are required for examination entrance.

- **Practical marks:** After completion of the experiment, the demonstrator will mark your report. Please follow up as to why you lost marks in your practical, especially in the first 3 practicals. These form the foundation for the rest of the practicals.
- **Practical tests:** There will be two practical tests on the laboratory work. You will not be required to make any measurements during the tests. Data will be given to you, based on what you have learned during the practicals sessions, and you will need to interpret the data and draw conclusions. There will also be a multiple choice section to the test.

2.7.10 Homework and Tutorial Sessions

Tutorial sessions are compulsory for **ALL** students. **No exemption will be given for the tutorials.** The marks received for the weekly tutorial tests all contribute to your final semester mark. Should problems occur with regards to attending a tutorial session, please submit a reason on the Google form "Absent from an assessment" mentioned previously.

Each week, you will have homework problems which will be similar to the problems which must be solved during the weekly tutorial test. Tutorial tests will be written at the end of each tutorial session. During the session, the tutor will be available to guide and assist students with the challenging homework problems. This does not mean that the tutor will be providing all the solutions to the problems; the problems are there for **YOU** to solve, and the tutor is merely a guide so that you can hone your problem solving techniques. **DO NOT CONSULT SOLUTIONS BEFORE YOU HAVE ATTEMPTED THE PROBLEM. THIS WILL NOT IMPROVE YOUR PROBLEM SOLVING SKILLS.**

2.8 Calculation of Marks

Semester Test 1	25%
Semester Test 2	30%
Online Test Average	10%
Weekly Tutorials	15%
Weekly Practicals	10%
Practical Tests 1	5%
Practical Tests 2	5%
Semester Mark	**100%**

Final Mark (SEM MARK + EXAM MARK)/2

Exemption from practicals: Should you have practical exemption, the calculation of your semester mark is the same as above, with the exception that the total mark (now out of 80) will be converted to a percentage.

2.9 Plagiarism

Plagiarism is when you present someone else's ideas – published or unpublished – as if they were your own. Other people's ideas may be contained in written text (journal articles, textbooks, etc.), visual text (graphics, photographs, etc.), multimedia products (websites, media productions, etc.), music (compositions, lyrics, etc.), and spoken text (speeches, lectures, etc.). Plagiarism is a serious offence and a student could be charged with misconduct which could lead to suspension from the University. For further information on plagiarism visit the following website: https://library.up.ac.za/plagiarism. Also refer to Section A19 in the General Academic Regulations.

Bringing this closer to home, if students are found copying from each other, or from work obtained from a student from a previous year (particularly for the practicals), this is also considered as plagiarism and is taken very seriously.

All students will need to make a declaration to the effect that all work submitted for assessment is their own work, and no assistance from any unauthorised source is allowed. You must familiarise yourself with the test and examination rules that can be found in the University of Pretoria Yearbook under General Rules and Regulations in Section G12 which can be found on the following link:

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The declaration will be in the form of a ClickUP Test, and the wording is as follows:

> *The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment*

You will need to agree to this statement.

3.1 Main Outcomes

At the end of this module you should be able to apply the principles learnt in the course to analyse and solve relevant problems in: Kinematics, Mechanics, Oscillations, Waves and Thermodynamics. More details for each section can be found on ClickUP.

3.2 Assessment Approach

Assessment will focus mainly on applying the theory to solve problems, on a similar level to those given as homework exercises. Marks will be allocated for clear, logical reasoning as well as for appropriate diagrams. Demonstrate clearly that you understand the physics that you use to answer the question. **This means that a numerically correct answer will not necessarily be awarded full marks.**

Knowledge of the theoretical aspects of the course (such as derivations from first principles) will also be assessed. Here strong emphasis will be placed on clear, logical reasoning.

It is essential that correct units are given in all solutions.

3.3 Summary of Chapters Covered

The following summary is based on the prescribed textbook by Serway and Jerwett. Please see ClickUP for more details on each section.

Topic	Chapter	Main Concepts
Measurement	1	Physical quantities, Models, Dimensional Analysis, Units, Order of magnitude, Significant figures
Motion in 1-dimension	2, 13	Position, Velocity, Acceleration, Kinematic equations under constant acceleration, Motion diagrams
Vectors	3	Coordinate systems, Vector and scalar quantities, Components of a vector, Unit vectors
Motion in 2-dimensions	4	Two-dimensional motion with constant acceleration, Projectile motion, Uniform circular motion, Relative velocity and acceleration
Laws of motion	5	Force, Newton's laws, Inertia, Mass, Gravitational force and weight, Frictional forces
Circular motion and resistive forces	6	Uniform circular motion (extended), Motion with resistive forces
Energy	7, 13	Systems and environments, Work done by constant and variable forces, Kinetic energy-work theorem, Potential Energy, Conservative forces
Conservation of energy	8	Isolated and non-isolated systems, Situations involving kinetic friction, Influence of non-conservative forces, Power
Linear momentum	9	Linear momentum for isolated and non-isolated systems, Collisions in 1- and 2-dimensions, Center of mass, Systems of particles
Rotation of a rigid object	10 and 11.1	Angular position, velocity, and acceleration, Constant angular acceleration, Angular and translational quantities, Torque, Moments of inertia, Rotational kinetic energy
Oscillatory motion	15	Motion of objects connected to springs, Simple harmonic motion, Energy of SHM, Pendulums
Wave motion	16	Propagation of a wave, Traveling waves, Speed of waves on strings, Sound waves, Doppler Effect
Superposition and standing waves	17	Interference, Standing waves, Boundary conditions, Air columns, Beats
Temperature	18	Zeroth law, Temperature scales, Thermometers, Thermal expansion of solids and liquids
First law of thermodynamics	19	Heat, Internal energy, Specific heat, Calorimetry, Latent heat, Work and heat in thermodynamic processes, First law, Applications

4.1 For e-learning support

Report a problem you experience to the Student Help Desk (Telephone: 012 420 3837, email:studenthelp@up.ac.za).
Approach the assistants at the help desks (adjacent to the Student Computer Laboratories in IT Building, NW2, CBT, etc).
Visit the open labs in the Informatorium Building to report problems at the offices of the Student Help Desk (Telephone: 012 420 3837, email:studenthelp@up.ac.za).

4.2 Safety in the evening: Green Route

From 18:00 till 06:00 Security Officers are available to escort you (on foot) to and from your residence or campus anywhere east of the Hatfield campus through to the LC de Villiers terrain. Departure point is at the ABSA ATM next to the Merensky Library.
Phone the Operational Management Centre if you need a Security Officer to accompany you from your residence to campus. See the table below for the contact telephone numbers for Campus Security Services.

4.3 Other student support services

FLY@UP	Think carefully before dropping modules (after the closing date for amendments or cancellation of modules). Make responsible choices with your time and work consistently. Aim for a good semester mark. Don't rely on the examination to pass.	www.up.ac.za/fly@up fly@up.ac.za	
Student Counselling Unit	Provides counselling and therapeutic support to students.	012 420 2333	
Student Health Services	Promotes and assists students with health and wellness.	012 420 5233 012 420 3423	
Careers Office	Provides support for UP students and graduates as they prepare for their careers.	careerservices@up.ac.za 012 420 2315	
Department of Security Services	24-hour Operational Management Centre 24-hour Operational Manager cell Crisis Line	012 420 2310 or 012 420 2760 083 654 0476 0800 006 428	
Department of Student Affairs	Enquiries concerning studies, accommodation, food, funds, social activities and personal problems.	012 420 2371/4001 Roosmaryn Building, Hatfield campus	

Centre for Sexualities, AIDS and Gender	Identifies and provides training of student peer counsellors.	012 420 4391
Disability Unit	Ensure an integrated and inclusive learning experience for students with disabilities.	012 420 2064
Fees and Funding	http://www.up.ac.za/enquiry http://www.up.ac.za/fees-and-funding	012 420 3111
IT Helpdesk	For student IT related queries.	012 420 3051 studenthelp@up.ac.za